

TRAZIA

Project Specification

“Help students spend less time organizing their semester and more time succeeding in it.”

Document Status: Draft | Version 1 (MVP)

Last Updated: July 16, 2026

1. Purpose

1.1 Problem Being Solved

- Students currently manually enter course assignments, exams, and grading information into spreadsheets or generic management tools.
- Trazia helps students map out their semester efficiently by transforming a syllabus into an organized academic plan, among other features.

1.2 Target Audience

- Primarily college students (undergraduate and graduate).
- Some high school students.

1.3 Motivation

- Built to solve a problem I've faced and that I find myself in struggling with at the start of every semester: manually uploading assignments and their due dates from 7+ classes every semester into an assignment tracker.
- The name is inspired by the Spanish verb “trazar,” meaning “to map out” or “to chart a course.”

2. Problem Statement

- Students manually enter assignments from syllabi or course schedules.
- Every syllabus is formatted differently and some course schedules are very long, making manual transcription error-prone.
- The process takes hours at the start of each semester and can kill the drive or motivation that students start with.
- Existing planning tools do not automate syllabus intake.

3. Solution Overview

Trazia is an AI-powered web application that extracts assignments, exams, grading information, deadlines, and relevant course details from uploaded course syllabi. Students review the extracted information before importing it into a personalized semester planner with grade tracking and study planning. This process saves hours before the semester starts and enables a smooth transition back into the academic world.

4. Goals

4.1 Functional Goals

- Upload syllabus
- Extract assignments
- Confirm extracted information
- Save courses
- Track assignments
- Track exams
- Grade calculator

4.2 Non-Functional Goals

- Easy to use
- Responsive across devices
- Fast imports
- Secure user accounts
- Accurate AI extraction

5. Target Users

- College students
- Graduate students
- Some high school students

6. MVP Scope (Version 1)

6.1 Authentication

- Create account
- Login/Logout

6.2 Courses

- Add course
- Course number
- Course name
- Course professor
- Course section

6.3 AI Import

- Upload PDF syllabus
- Extract assignments, exams, grading categories
- Confirmation screen

6.4 Dashboard

- Upcoming assignments
- Sort by due date
- Assignment status
- Edit assignments

6.5 Later Versions

- Calendar view
- Notifications
- Canvas integration
- Study planner

7. Features

7.1 Authentication

- Email login

7.2 Dashboard

- Upcoming assignments
- Color coding
- Due dates
- Office hours

7.3 Assignments

- Status
- Progress
- Score
- Points possible

7.4 Exams

- Study hours
- Days before exam

7.5 Grade Calculator

- Current grade
- Predicted grade

8. Technology Stack

Layer	Technology
Frontend	Next.js, React, Tailwind CSS
Backend	FastAPI, Python
Database	PostgreSQL (Supabase)
Deployment	Vercel (frontend), Render (backend)
AI	OpenAI API

9. Version Roadmap

Version	Codename	Focus
Version 1	MVP	Core syllabus upload, AI extraction, dashboard, grade calculator
Version 2	Planner	Full semester planning tools
Version 3	Study Assistant	Study scheduling and task generation
Version 4	AI Assistant	Conversational / intelligent recommendations
Version 5	Integrations	Canvas, calendars, and other stretch goals

10. Stretch Goals

- Canvas integration
- Notifications
- AI-driven scheduling
- Workload prediction
- Time estimation
- Dark mode/themes

11. Success Criteria

The MVP will be considered successful when the following criteria are met:

☐	Success Criterion
☐	A new user can upload a syllabus.
☐	AI extracts assignments correctly.
☐	User confirms import.
☐	Dashboard displays assignments.
☐	Grade calculator works.
☐	Data persists between sessions.

12. Future Vision

Trazia aims to become an AI-powered academic planning platform rather than simply an assignment tracker. Future versions will provide personalized study planning, workload forecasting, calendar synchronization, and intelligent recommendations that help students stay organized throughout the semester.